

Anirudh S Chakravarthy

✉ anirudh.s.chakravarthy@gmail.com • 🌐 anirudh-chakravarthy.github.io
in anirudh-chakravarthy

Experience

Amazon AGI SF Lab

San Francisco, CA

Member of Technical Staff

Apr 2025 – Present

- Helped deliver Nova Act on AWS – a browser-use agent with comparable performance to Claude Sonnet 4.5 on multimodal understanding, web navigation, and UI understanding benchmarks
- Owned end-to-end post-training for knowledge work benchmarks (GDPVal and Finance Agent), including synthetic data generation, RL/SFT, evaluation, and failure analysis; improving internal benchmarks by 20%
- Led technical initiative to introduce the human takeover capability, enabling handoff for credentials, captchas, and approval for financial transactions, with collaboration on SFT trajectory collection, vendor gym creation and task generation, eval creation based on product shape, and engineering to develop model-harness contract
- Owned human takeover capability research: data generation strategy and ingestion, evaluation and gym infrastructure, small-scale post-training recipe, harness iteration, and upstreaming into large model run
- Led synthetic data generation research and productionized exploration-driven proposer to stitch atomic trajectories into complex tasks based on sequential exploration, improving internal benchmarks by 10% after SFT
- Led self-play RL research to train a task proposer capable of generating tasks at the frontier of model's capability; demonstrating step-improvements on task complexity and quality and improving internal benchmarks by 5%

Cruise LLC

San Francisco, CA

Senior Applied Scientist

Feb 2023 – Mar 2025

- Identified critical on-road misclassifications of gesture and pedestrian posture; led algorithmic and onboard effort for camera-lidar fusion to improve gesture recognition by 10% and pedestrian-falling classification by 15%
- Owned development of new perception attributes for children, wheelchair, etc: data miners to identify logs of interest, vendor collaboration for labeling, data-mix experiments; achieved > 85% recall while avoiding regressions
- Led collaborations with behavior planning for system design and end-to-end integration (e.g: responding to police-motorcycles blockages, yielding for children and strollers), improving caution around pedestrians by 5×
- Introduced off-board foundation models to identify failure modes of deployed model, source high-quality training data, and continually improve production model; consistently achieved >5% improvements using mined data
- Developed a task-incremental optimizer to dynamically balance task gradients while maintaining performance on old tasks, achieving SOTA on VTAB, DriveLM, and Waymo Motion Dataset; accepted at **NeurIPS'25**
- Implemented a multi-task learning framework to adaptively balance task weights during training; improved precision on school bus classification by 7% and delivery vehicle classification by 3% over production model

CMU Argo AI Center

Pittsburgh, PA

Research Collaborator (mentor: Dr. Deva Ramanan)

Nov 2021 – Dec 2022

- Introduced Open-World LiDAR Panoptic Segmentation, enabling segmentation of objects unseen during training
- Proposed a panoptic segmentation method combining LiDAR semantic segmentation with non-learned instance grouping; improved closed-set SOTA by 3% Panoptic Quality, open-world methods by 20% Unknown Quality

Education

Carnegie Mellon University, School of Computer Science

Pittsburgh, PA

Master of Science in Computer Vision (Robotics Institute)

GPA: 4.14/4

Birla Institute of Technology and Science

Pilani, India

Bachelor of Engineering in Computer Science (Minor: Data Science)

GPA: 9.55/10; Department Rank: 9/153; Top 2% Merit Scholarship

Publications

Anirudh S Chakravarthy, Shuai Kyle Zheng, Xin Huang, Sachithra Hemachandra, Xiao Zhang, Yuning Chai, and Zhao Chen. PROFIT: A Specialized Optimizer for Deep Fine Tuning. In *Neural Information Processing Systems (NeurIPS)*, 2025.

Anirudh S Chakravarthy, Meghana Ganesina, Peiyun Hu, Laura Leal-Taixe, Shu Kong, Deva Ramanan, and Aljosa Osep. Lidar Panoptic Segmentation in an Open World. In *International Journal of Computer Vision (IJCV)*, 2024.

Donglai Wei et al. YouMVOS: An Actor-centric Multi-shot Video Object Segmentation Dataset. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.